

RAHUL MITRA

AI ENGINEER PROFILE

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PROFILE SUMMARY

Early-career AI engineer focused on applied model systems, RAG, agentic workflows, NLP, computer vision, and AI-assisted creation pipelines. Built AI products across maritime, healthcare, sustainability, education, geospatial monitoring, and 3D media workflows.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Artificial Intelligence; 3D Computer Vision; Software Engineering; Simulation; Statistics; Operations Research; Linear Algebra; Probability

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERNSHIP EXPERIENCE

Data Analytics and Product Management Intern, Abbott Laboratories Singapore Pte Ltd

Jan 2026 - Present

- Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules.
- Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.
- Offered a follow-on 7-month Data Engineer, Operational AI Systems contract, extending work across manufacturing analytics, deployment reliability, and operational AI workflows.

Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Awarded the S\$20,000 Sparks Community Innovation Fund at the 2025 Community Hackathon to continue Churp for Tengah community gardening operations.
- Built product workflows for transparent plot allocation, resident feedback, and civic service delivery while conducting user testing to refine product requirements.

Independent Bug Bounty Researcher, YesWeHack Platform

May 2024 - Present

- Assessed GovTech GBBP12/13, Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and authorization weaknesses.
- Used Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts to automate endpoint discovery, payload delivery, and evidence collection for responsible disclosure reports.

SELECTED PROJECTS AND COMPETITIONS

OnTheSpectrum - Codex and Blender MCP 3D Pipeline

May 2026

- Built an innovative local-first Blender-to-Three.js asset pipeline using Codex, Blender MCP, generated GLB assets, metadata, previews, playable world QA, and Remotion video creation workflows.

Maritime Deficiency Severity Forecasting

2025

- Led model training for a maritime inspection NLP system using a fine-tuned 109M-parameter BERT model, DNN layers, gradient clipping, and ASPIRE 2A supercomputing.

AgeWellLah.AI - NUS HealthHack 2025

2025

- Engineered an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, Flask, OAuth2-secured patient profiles, and PDPA-aware design choices.

SmartExam - GovTech x NTUPC Product Hackathon

2025

- Built a multi-agent autonomous exam generation system with RAG, Next.js, source ingestion, question generation, and education workflow automation.

EthosLens - LifeHack 2025

2025

- Developed backend workflows for an AI research engine that scores brand ethics and sustainability practices using FastAPI, Python, LangChain, and automated evidence retrieval.

NVIDIA Disaster Risk Monitoring with Satellite Imagery

2023

- Completed hands-on geospatial AI training covering satellite imagery, U-Net semantic segmentation, disaster-risk assessment, and GPU-accelerated model workflows.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Python; BERT; DNN; RAG; AI Agents; OpenAI GPT-4; Gemini; Vector Search; SEALION; ASPIRE 2A; Codex; Blender MCP; Remotion; Computer Vision; Model Evaluation.

- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.